greatest common divisor of m and n. It is now easy to understand that $\sqrt{\Delta}$ is rational if and only if both m and n are perfect squares. Also, $\sqrt{\Delta}$ is irrational if and only if at least one of m and n is not a perfect square.

We are familiar with irrational numbers of the type $p+\sqrt{q}$ where p and q are rational numbers and $\sqrt{q}$ is irrational. Such numbers are called **surds**. As in the case of imaginary roots, we can prove that if $p+\sqrt{q}$ is a root of a polynomial, then $p-\sqrt{q}$ is also a root of the same polynomial equation, when all the coefficients are rational numbers. Though this is true for polynomial equation of any degree and can be proved using the technique used in the proof of imaginary roots, we state and prove this only for a quadratic equation in Theorem 3.3.

Before proving the theorem, we recall that if a and b are rational numbers and c is an irrational number such that $a+bc$ is a rational number, then b must be 0; further if $a+bc=0$, then a and b must be 0.

For instance, if $a+b\sqrt{2}\in\mathbb{Q}$, then b must be 0, and if $a+b\sqrt{2}=0$ then $a=b=0$. Now we state and prove a general result as given below.

Theorem 3.3

Let p and q be rational numbers such that $\sqrt{q}$ is irrational. If $p+\sqrt{q}$ is a root of a quadratic equation with rational coefficients, then $p-\sqrt{q}$ is also a root of the same equation.

Proof

We prove the theorem by assuming that the quadratic equation is a monic polynomial equation. The result for non-monic polynomial equation can be proved in a similar way.

Let p and q be rational numbers such that $\sqrt{q}$ is irrational. Let $p+\sqrt{q}$ be a root of the equation $x^2+bx+c=0$ where b and c are rational numbers.

Let α be the other root. Computing the sum of the roots, we get

$$\alpha+p+\sqrt{q} = -b$$

and hence $\alpha+\sqrt{q}=-b-p\in\mathbb{Q}$. Taking $-b-p$ as s, we have $\alpha+\sqrt{q}=s$.

This implies that

$$\alpha = s-\sqrt{q}.$$

Computing the product of the roots, we get

$$(s-\sqrt{q})(p+\sqrt{q}) = c$$

and hence $(sp-q)+(s-p)\sqrt{q}=c\in\mathbb{Q}$. Thus $s-p=0$. This implies that $s=p$ and hence we get $\alpha=p-\sqrt{q}$. So, the other root is $p-\sqrt{q}$. ■

Remark

The statement of Theorem 3.3 may seem to be a little bit complicated. We should not be in a hurry to make the theorem short by writing "**for a polynomial equation with rational coefficients, irrational roots occur in pairs**". This is not true.

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